

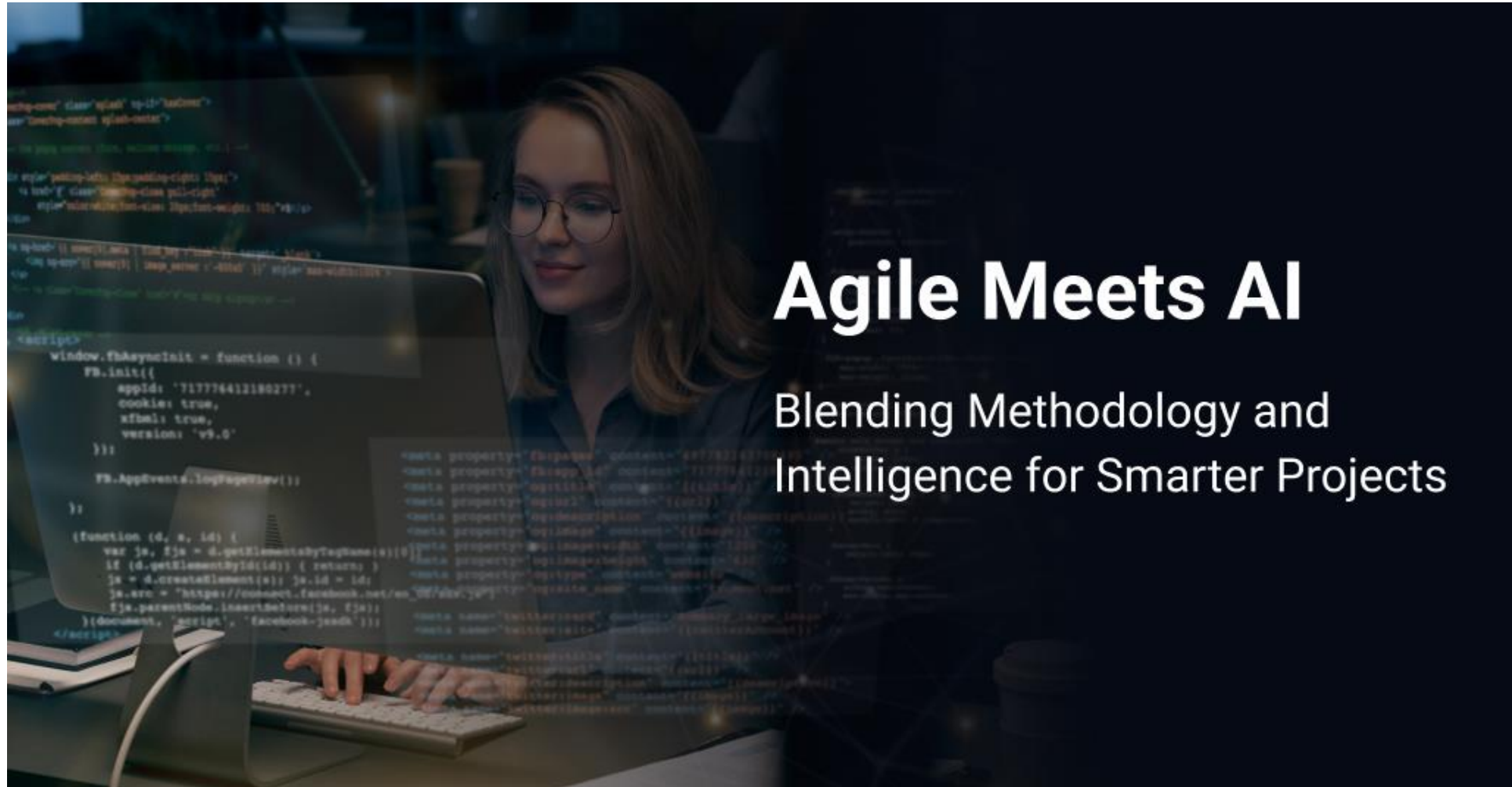
Lecture 3

AI-CR-503

Digital Technology Program

Madan Bhandari University of Science and Technology (MBUST)

UNIT 3 Agile methods



Introduction to Agile

- Agile means flexibility, adaptability, and iterative approach to work or project management.
- Agile emphasizes collaboration.
- Agile philosophy is human-centred in nature.

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- Agile manifesto consists of 12 principles.
- Agile principles have been implemented in Software Engineering (SE) to address the complexity of modern software developments.

Scrum and Kanban

- Scrum is an agile framework, where team works in short and fixed iterations called sprints on defined time. The sprints can be 1-4 weeks long with specific scope. The project is delivered in incremental amounts until the end.
- Kanban is the continuous methods without defined time. The work is displayed on the kanban board and move through the stages. The project is delivered at the end.

Agile principles (12)

1. Customer satisfaction through early and continuous delivery of valuable software.
2. Welcome changing requirements at any development stage.
3. Frequent working software delivery.
4. Close co-operation between business people and developers.
5. Foster motivation of individuals within the team.
6. Preference for face-to-face interaction.
7. Working software as a primary measure of progress.

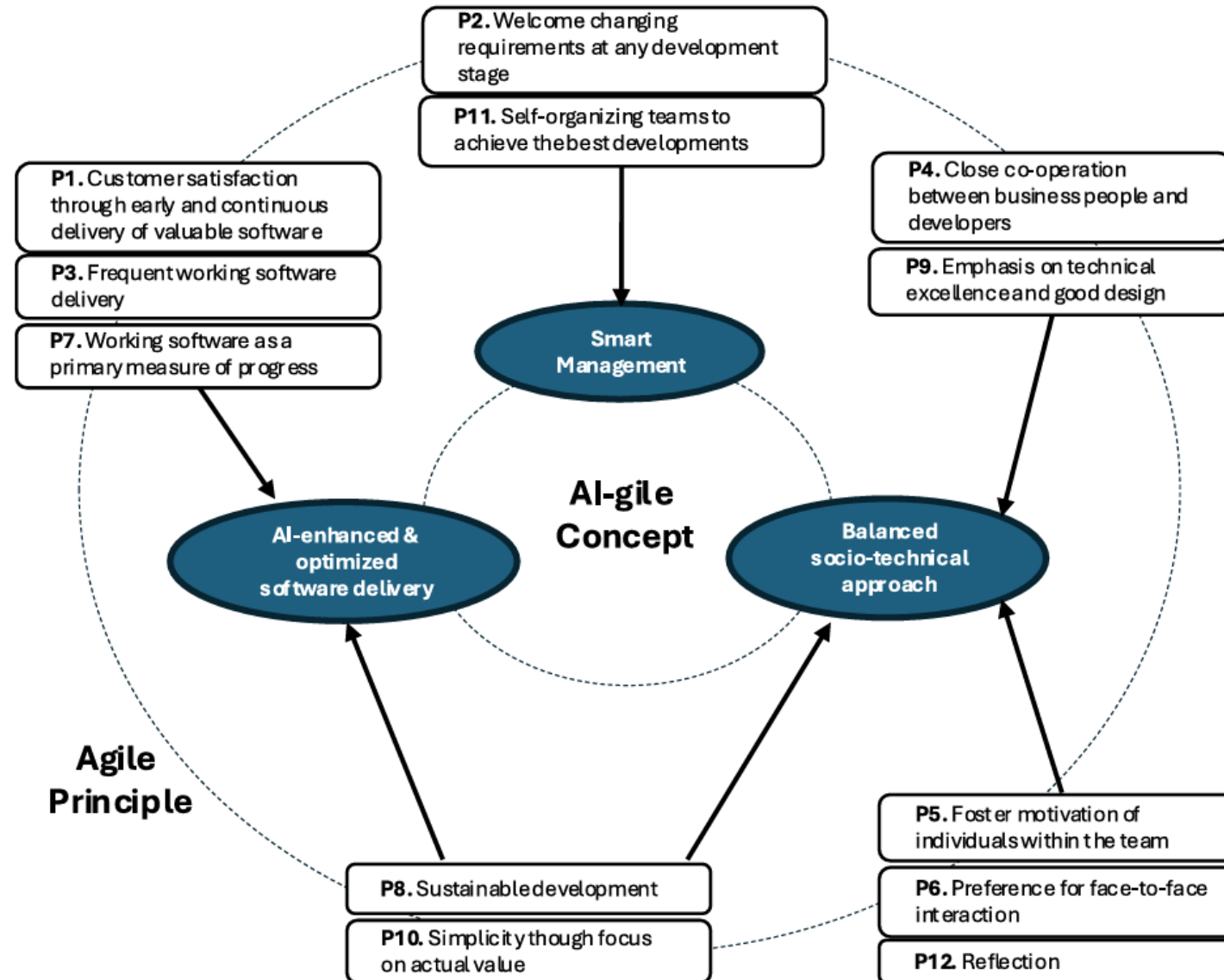
8. Sustainable development.
9. Emphasis on technical excellence and good design.
10. Simplicity though focus on actual value.
11. Self-organizing teams to achieve the best developments.
12. Reflection.

Agile methods in AI

The rise of AI has impacted different disciplines and domains, including software engineering (SE). Agile methods in AI try to incorporate Agile principles into AI applications development.

- a) AI enhanced and optimized software delivery
- b) Smart management
- c) Balanced socio-technical approach

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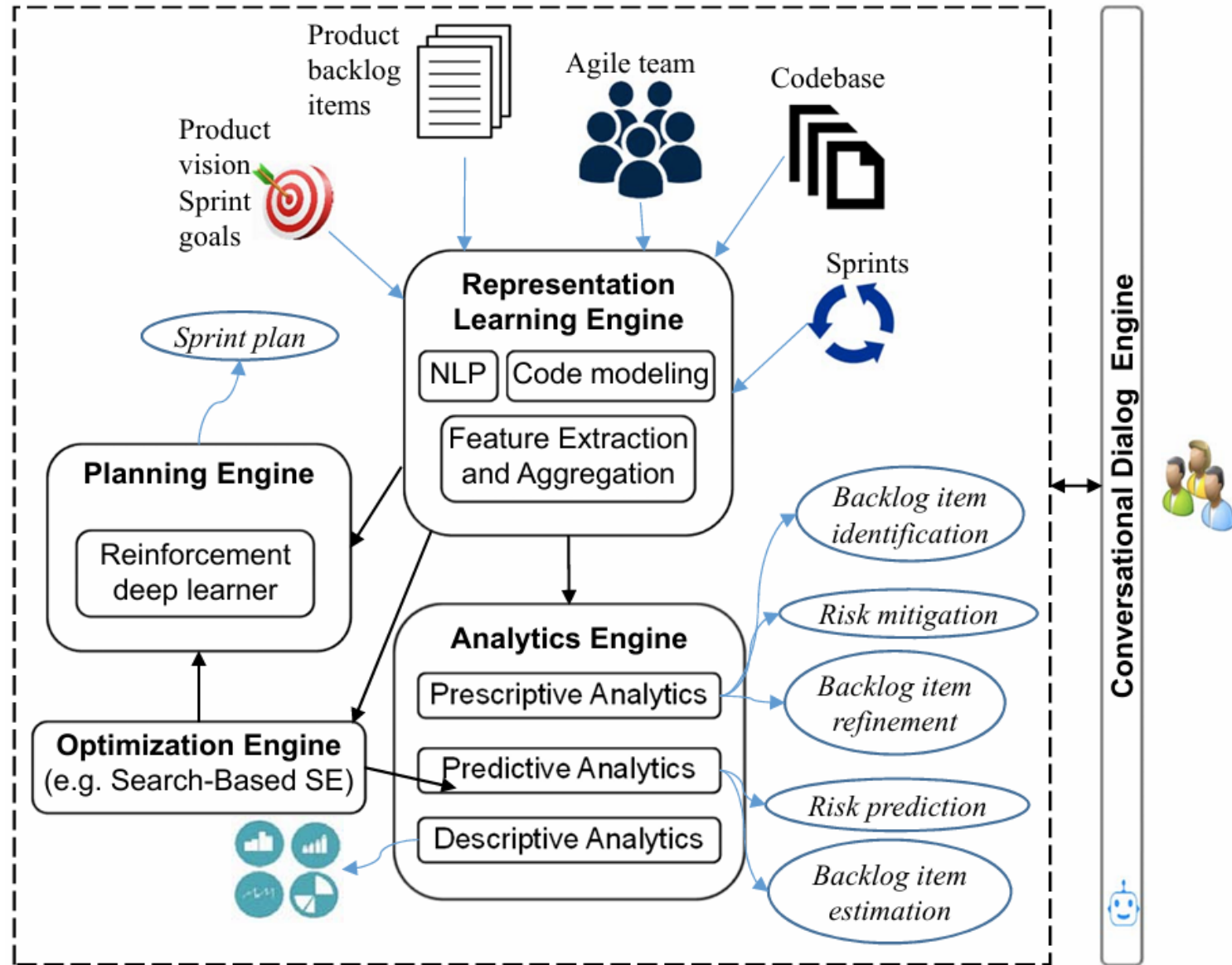


Difference between Agile and AI Agile

S.N.	Agile	AI Agile
1	Planning is based on team experiences.	AI analyze previous datasets.
2	Feedbacks from stakeholders.	Model performance gives feedbacks.
3	Respond to changing environment.	Respond to evolving data and algorithms.
4	Human provide updates on project development.	AI tools highlights the progress.

Advantages of Agile method in AI

- AI can significantly improve the practice of agile project management by considering three engine.
 - a) Analytics engine
 - b) Planning engine
 - c) Optimization engine.
- These engines depends on representation learning engine to learn and generate project data that are easy to process.



AI powered agile project assistant

1. Backlog item identification
2. Risk mitigation
3. Backlog item refinement
4. Risk prediction
5. Backlog item estimation

Applications of Agile method in AI

- Finance: Predictive analytics.
- Smart agriculture: Predicts crop yields or disease outbreak.
- Healthcare diagnostics: Use imaging data to classify disease.
- Natural language processing: Chatbots and translation systems.

Thank you.